Lecture 7. Optimization

Golden section, Newton, line search, Fisher scoring, GLM and IRLS, Gauss–Newton, gradient and coordinate descent

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We consider $\min_{x\in\mathbb{R}^d} f(x)$; maximization is minimization of $-f$.

7.1 Golden section search

For $d = 1$, let f be continuous and **strictly unimodal** on $[a, b]$: $f(\lambda x+(1-\lambda)y) < \max\{f(x), f(y)\}$ for $x \neq y$, $\lambda \in (0, 1)$. Keep $a < x_1 < b$ with $f(x_1) < \min\{f(a), f(b)\}$, so the minimum is in (a, b), and place x_0 in the longer of (a, x_1), (x_1, b); say $a < x_0 < x_1$.

- If $f(x_0) > f(x_1)$, the minimum is in (x_0, b): set $(a, x_1, b) \leftarrow (x_0, x_1, b)$.
- Otherwise it is in (a, x_1): set $(a, x_1, b) \leftarrow (a, x_0, x_1)$.

For equally good outcomes and self-similarity, require $b - x_0 = x_1 - a$ and that the interior point divide the new interval in the same proportion. With $\alpha = (x_0 - a)/(b - a)$, $y = (x_1 - x_0)/(b - a)$: $\alpha + y + \alpha = 1$ and $1 : \alpha = (y + \alpha) : y$, so

$$\alpha = \frac{3-\sqrt{5}}{2}, \qquad 1-\alpha = \frac{\sqrt{5}-1}{2} \approx 0.618,$$

the golden section. Each step shrinks the interval by 0.618 with one evaluation: linear, but guaranteed to find the global minimum of a unimodal function.

7.2 Newton's method

At a minimizer $\nabla f(x^\star) = 0$. Newton–Raphson on ∇f minimizes the local quadratic model

$$f(x) \approx f(x^{(t)}) + \nabla f(x^{(t)})^\top (x - x^{(t)}) + \tfrac{1}{2}(x - x^{(t)})^\top \nabla^2 f(x^{(t)})(x - x^{(t)}),$$

giving the **naive Newton** iteration

$$x^{(t+1)} = x^{(t)} - [\nabla^2 f(x^{(t)})]^{-1} \nabla f(x^{(t)}).$$

It converges quadratically near a minimizer with positive definite Hessian, but it is not a descent method: it can converge to a maximum or saddle, or diverge (try $f = \sin$).

7.3 Practical Newton: positive definite approximations and line search

Motivation. A positive definite Hessian substitute and a line search keep Newton's fast local convergence while guaranteeing descent.

Use a positive definite $H^{(t)}$ and a step size s:

$$x^{(t+1)} = x^{(t)} + s\,\Delta x^{(t)}, \qquad \Delta x^{(t)} = -[H^{(t)}]^{-1} \nabla f(x^{(t)}).$$

Proposition 7.1 (Descent direction)

If $H^{(t)}$ is positive definite and $\nabla f(x^{(t)}) \neq 0$, then $f(x^{(t)} + s\Delta x^{(t)}) < f(x^{(t)})$ for all sufficiently small $s > 0$.

Proof. By Taylor, $f(x^{(t)} + s\Delta x^{(t)}) - f(x^{(t)}) = -s\,\nabla f^\top [H^{(t)}]^{-1} \nabla f + o(s)$ with a positive quadratic form. ∎

($\{\nabla f^\top H^{-1} \nabla f\}^{1/2}$ is the **Newton decrement**.) For strictly convex f the true Hessian works (**damped Newton**), still with a line search for global convergence; $H^{(t)} = I$ is gradient descent.

Line search. Choose s exactly (minimize $f(x+s\Delta x)$, e.g. golden section) or approximately by step halving or **backtracking** (Armijo): with $\alpha \in (0, \frac{1}{2})$, $\beta \in (0,1)$, start at $s=1$ and set $s \leftarrow \beta s$ while

$$f(x+s\Delta x) > f(x) + \alpha s \nabla f(x)^\top \Delta x.$$

This accepts s once f is below a line with slope a fraction α of the initial slope, which holds for small s by Proposition 7.1; it costs only function evaluations. A common slip: with the trial point $x_s = x + s\Delta x$ formed first, the bound is $f(x) + \alpha \nabla f(x)^\top (x_s - x)$, which already contains s; writing $\alpha s \nabla f(x)^\top (x_s - x)$ counts s twice and is too lenient. (For IRLS, the step-s trial point is the weighted least squares fit to $z = \eta + sW^{-1}(y-\mu)$, which is linear in z and returns $\beta + s\Delta\beta$.)

7.4 Fisher scoring

For maximum likelihood, $f = -\ell(\theta)$. **Fisher scoring** replaces $-\nabla^2 \ell$ by the expected information

$$I(\theta) = E[-\nabla^2 \ell(\theta)] = E[\nabla \ell(\theta) \nabla \ell(\theta)^\top] \succeq 0,$$

(equality when differentiation and expectation interchange), giving

$$\theta^{(t+1)} = \theta^{(t)} + s[I(\theta^{(t)})]^{-1} \nabla \ell(\theta^{(t)}).$$

Example 7.1 (Logistic regression)

$y_i \sim \mathrm{Ber}(p_i)$ with $\operatorname{logit} p_i = \eta_i = x_i^\top \beta$. Then

$$\ell(\beta) = \sum_i \{y_i x_i^\top \beta - \log(1 + e^{x_i^\top \beta})\}, \qquad \nabla \ell = X^\top (y - p),$$

$$-\nabla^2 \ell = X^\top W X, \qquad W = \mathrm{diag}\{p_i(1-p_i)\}.$$

The Hessian is free of y, so $I(\beta) = -\nabla^2 \ell$ and Newton = Fisher scoring. Rearranged,

$$\beta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} z^{(t)}, \qquad z^{(t)} = X\beta^{(t)} + s(W^{(t)})^{-1}(y - p^{(t)}),$$

a weighted least squares fit to the **working responses** $z^{(t)}$: **iteratively reweighted least squares** (IRLS). Any weighted least squares solver fits logistic regression.

Example 7.2 (Poisson regression)

$y_i \sim \mathrm{Poi}(\lambda_i)$, $\log \lambda_i = x_i^\top \beta$: $\ell(\beta) = \sum_i \{y_i x_i^\top \beta - e^{x_i^\top \beta} - \log y_i!\}$, $\nabla \ell = X^\top (y - \lambda)$, $-\nabla^2 \ell = X^\top W X$, $W = \mathrm{diag}(\lambda_i)$; again Newton = Fisher scoring = IRLS with $z = X\beta + W^{-1}(y - \lambda)$.

7.5 Generalized linear models

An **exponential dispersion family** has density

$$p(y \mid \eta, \phi) = \exp\left\{\frac{y\eta - b(\eta)}{a(\phi)} + c(y, \phi)\right\},$$

with natural parameter η, mean $\mu = b'(\eta)$, variance $b''(\eta)a(\phi)$. A **GLM** sets $g(\mu) = x^\top \beta$; the **canonical link** $g = (b')^{-1}$ makes $\eta = x^\top \beta$.

family	canonical link	variance function
normal	$\eta = \mu$	1
Poisson	$\eta = \log \mu$	μ
binomial	$\eta = \log \frac{\mu}{1-\mu}$	$\mu(1-\mu)$
gamma	$\eta = \mu^{-1}$	μ^2
inverse Gaussian	$\eta = \mu^{-2}$	μ^3

Proposition 7.2

With $a(\phi) \equiv 1$, $\mu_i = g^{-1}(x_i^\top \beta)$ and $\sigma_i^2 = b''(\eta_i)$:

$$\nabla \ell(\beta) = \sum_i \frac{(y_i - \mu_i)}{\sigma_i^2 g'(\mu_i)} x_i, \qquad I(\beta) = \sum_i \frac{1}{\sigma_i^2 g'(\mu_i)^2} x_i x_i^\top = X^\top W X.$$

Fisher scoring is IRLS with weights $w_i = 1/\{\sigma_i^2 g'(\mu_i)^2\}$. For the canonical link, $-\nabla^2 \ell = I(\beta)$, so Newton = Fisher scoring = IRLS and $-\ell$ is convex.

Proof. $\ell_i = y_i \eta_i - b(\eta_i)$, so $\partial \ell_i / \partial \eta_i = y_i - \mu_i$. As $d\mu = b''(\eta) d\eta = \sigma^2 d\eta$ and $d\mu = x^\top d\beta / g'(\mu)$, $\partial \eta_i / \partial \beta = x_i / \{\sigma_i^2 g'(\mu_i)\}$: the score. Differentiating again gives $-x_i x_i^\top / \{\sigma_i^2 g'(\mu_i)^2\}$ plus a term proportional to $y_i - \mu_i$ (mean 0): $I(\beta)$. For the canonical link $\partial \eta_i / \partial \beta = x_i$ and $\nabla^2 \ell = -\sum_i b''(\eta_i) x_i x_i^\top = -I(\beta)$, free of y_i ($\sigma_i^2 g'(\mu_i) = 1$). ■

For non-canonical links (e.g. probit) Fisher scoring and Newton differ and the likelihood need not be concave (probit happens to be log-concave). R's `glm` uses Fisher scoring.

Example 7.3 (Probit regression)

$y_i \sim \text{Ber}(p_i)$, $p_i = \Phi(\eta_i)$, $\eta_i = \beta_0 + \beta_1 x_i$; write $\Phi_i = \Phi(\eta_i)$, $\phi_i = \phi(\eta_i)$, $\mathbf{x}_i = (1, x_i)^\top$. Then

$$\ell(\beta) = \sum_i \{y_i \log \Phi_i + (1 - y_i) \log(1 - \Phi_i)\}, \qquad \nabla \ell = \sum_i \frac{(y_i - \Phi_i)\phi_i}{\Phi_i(1 - \Phi_i)} \mathbf{x}_i.$$

With $\phi'(z) = -z\phi(z)$,

$$\nabla^2 \ell = \sum_i f_i \, \mathbf{x}_i \mathbf{x}_i^\top, \qquad f_i = -y_i \frac{\phi_i(\eta_i \Phi_i + \phi_i)}{\Phi_i^2} - (1 - y_i) \frac{\phi_i \{\phi_i - \eta_i(1 - \Phi_i)\}}{(1 - \Phi_i)^2},$$

which depends on y_i (non-canonical link). With $Ey_i = \Phi_i$ the Fisher information is $X^\top W X$, $w_i = \phi_i^2 / \{\Phi_i(1 - \Phi_i)\}$, as in Proposition 7.2 ($\sigma_i^2 = \Phi_i(1 - \Phi_i)$, $g'(\mu_i) = 1/\phi_i$). The IRLS step is

$$\beta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} z^{(t)}, \qquad z_i^{(t)} = \eta_i^{(t)} + \frac{y_i - \Phi_i^{(t)}}{\phi_i^{(t)}}.$$

7.6 Nonlinear least squares: Gauss–Newton

(Gauss located Ceres in 1801 by least squares.) Minimize

$$f(\beta) = \tfrac{1}{2} \sum_{i=1}^{n} \{y_i - \mu_i(\beta)\}^2,$$

e.g. a logistic growth curve $\mu(x; \beta) = \beta_3 / (1 + e^{-\beta_1 - \beta_2 x})$. With $J(\beta)$ the $n \times p$ Jacobian with rows $\nabla \mu_i(\beta)^\top$,

$$\nabla f = -J^\top (y - \mu), \qquad \nabla^2 f = J^\top J - \sum_i (y_i - \mu_i) \nabla^2 \mu_i.$$

Gauss–Newton keeps only the positive semidefinite $J^\top J$:

$$\beta^{(t+1)} = \beta^{(t)} + s(J^\top J)^{-1} J^\top (y - \mu(\beta^{(t)})),$$

i.e. each step regresses the residuals on J. It suits small residuals or nearly linear μ, and is Fisher scoring for $y_i \sim N(\mu_i(\beta), \sigma^2)$, as $E[-\nabla^2 \ell] \propto J^\top J$.

7.7 Gradient descent

$$x^{(t+1)} = x^{(t)} - \gamma_t \nabla f(x^{(t)}).$$

$\Delta x = -\nabla f / \|\nabla f\|$ minimizes the linear model $\nabla f^\top \Delta x$ over $\|\Delta x\|_2 \le 1$ (steepest descent); the model is unbounded, so a step size is always needed. Steps are cheap and Hessian-free, attractive at large scale, but the method zigzags, converges slowly, and needs smooth f.

Theorem 7.3

Let f be convex and differentiable with L-Lipschitz gradient, $\|\nabla f(x) - \nabla f(y)\|_2 \le L\|x - y\|_2$, and let f attain its minimum $p^\star$ at $x^\star$. With constant step $\gamma \in (0, 1/L]$,

$$f(x^{(t)}) - p^\star \le \frac{\|x^{(0)} - x^\star\|_2^2}{2\gamma t}.$$

Proof. The Lipschitz gradient gives $f(y) \le f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2$; with $x^+ = x - \gamma \nabla f(x)$, $\gamma L \le 1$: $f(x^+) \le f(x) - \frac{\gamma}{2}\|\nabla f(x)\|^2$. Convexity gives $f(x) \le p^\star + \nabla f(x)^\top (x - x^\star)$, so

$$f(x^+) - p^\star \le \nabla f(x)^\top (x - x^\star) - \frac{\gamma}{2}\|\nabla f(x)\|^2 = \frac{1}{2\gamma}(\|x - x^\star\|^2 - \|x^+ - x^\star\|^2).$$

Summing telescopes: $\sum_{k=1}^t \{f(x^{(k)}) - p^\star\} \le \frac{1}{2\gamma}\|x^{(0)} - x^\star\|^2$, and the left side is $\ge t\{f(x^{(t)}) - p^\star\}$ as $f(x^{(k)})$ is nonincreasing. ∎

The $O(1/t)$ rate is *sublinear*; strong convexity gives linear convergence, never quadratic.

Example 7.4

- Least squares $f(\beta) = \frac{1}{2}\|X\beta - y\|^2$: $L = \sigma_{\max}(X)^2$, so $\gamma \le 1/\sigma_{\max}(X)^2$.
- Logistic regression: $L = \frac{1}{4}\|X^\top X\|_2$ as $p(1 - p) \le \frac{1}{4}$. The minimum may not exist (separable data); a ridge penalty $\frac{\rho}{2}\|\beta\|^2$ fixes this.
- Poisson regression: $-X^\top (y - e^{X\beta})$ is not globally Lipschitz, so a constant step can fail and line search is needed. An L-Lipschitz gradient forces $v^\top \nabla^2 f(\beta) v \le L\|v\|^2$, but if $x_j^\top v > 0$ for some j,

$$v^\top \nabla^2 f(tv) v = \sum_i e^{t x_i^\top v} (x_i^\top v)^2 \ge e^{t x_j^\top v} (x_j^\top v)^2 \to \infty \quad (t \to \infty).$$

Example 7.5 (Multinomial logistic regression)

With K classes, dummy-coded $y_i \in \{0, 1\}^K$ and $\log(p_{ik}/p_{iK}) = x_i^\top \beta_k$ ($k < K$, $\beta_K = 0$), $p_{ik} = e^{x_i^\top \beta_k}/(1 + \sum_{l<K} e^{x_i^\top \beta_l})$. With $B = [\beta_1, \ldots, \beta_{K-1}]$, Y_{-K} the first $K - 1$ columns of Y

and $P = (p_{ik})_{n\times(K-1)}$,

$$\ell(B) = \sum_i \sum_k y_{ik} \log p_{ik} = \operatorname{tr}(Y_{-K}^\top X B) - \sum_i \log\Big(1 + \sum_{l<K} e^{x_i^\top \beta_l}\Big),$$

$$\nabla_{\beta_k} \ell = \sum_i x_i (y_{ik} - p_{ik}), \qquad \text{so} \qquad B^{(t+1)} = B^{(t)} + \gamma X^\top (Y_{-K} - P^{(t)})$$

is gradient ascent ($K = 2$: logistic regression), $O(npK)$ per step with no $p(K-1)$-dimensional Hessian; the Hessian is bounded by $\frac{1}{2}\|X\|_2^2$, so a constant step works.

Example 7.6 (Quantiles by optimization)

The α-quantile of X (cdf F, density $f > 0$) solves $F(x) = \alpha$; Newton gives $x^{(t+1)} = x^{(t)} - \{F(x^{(t)}) - \alpha\}/f(x^{(t)})$, i.e. Newton minimization of ψ with $\psi' = F - \alpha$, $\psi'' = f > 0$. As $F(x) = E[\mathbf{1}\{X \le x\}]$,

$$\psi(x) = E\big[(x - X)\mathbf{1}\{X \le x\}\big] - \alpha x = E\big[\rho_\alpha(X - x)\big] + \text{const}, \qquad \rho_\alpha(u) = u\{\alpha - \mathbf{1}(u < 0)\},$$

the expected **check loss** of quantile regression, convex. Gradient descent $x^{(t+1)} = x^{(t)} - \gamma_t\{F(x^{(t)}) - \alpha\}$ needs no density; if only sampling from F is easy, use the unbiased one-sample gradient,

$$x^{(t+1)} = x^{(t)} - \gamma_t(\mathbf{1}\{X_t \le x^{(t)}\} - \alpha), \qquad X_t \overset{\text{iid}}{\sim} F,$$

with $\sum_t \gamma_t = \infty$, $\sum_t \gamma_t^2 < \infty$ (e.g. $\gamma_t = c/t$): this *stochastic gradient* (Robbins–Monro) algorithm converges to the quantile a.s. Alternatively, the sample α-quantile of N draws has standard error $\sqrt{\alpha(1-\alpha)/N}/f(x_\alpha)$.

7.8 Coordinate descent

Minimize over one coordinate (or block) at a time, cycling:

$$x_j^{(t+1)} = \arg\min_{x_j} f(x_1^{(t+1)}, \ldots, x_{j-1}^{(t+1)}, x_j, x_{j+1}^{(t)}, \ldots, x_d^{(t)}).$$

Each one-dimensional problem is often easy (sometimes closed form), and the objective decreases monotonically: the optimization analogue of Gauss–Seidel. For nondifferentiable convex f it can stick at a non-optimal "corner" where no single coordinate move helps.

7.9 Selected exercises

Motivation. Linear convergence under strong convexity, termination of backtracking, and coordinate descent as Gauss–Seidel.

Exercise 7.1 (Linear convergence under strong convexity)

Let f be differentiable with L-Lipschitz gradient and μ-*strongly convex*, $\mu > 0$:

$$f(y) \ge f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|y - x\|^2 \quad \text{for all } x, y,$$

and let $x^\star$ minimize f, $p^\star = f(x^\star)$.
(a) Show that $\|\nabla f(x)\|^2 \ge 2\mu\{f(x) - p^\star\}$ for all x.

(b) Show that gradient descent with $\gamma = 1/L$ satisfies $f(x^{(t)}) - p^\star \le (1-\mu/L)^t\{f(x^{(0)}) - p^\star\}$.
(c) For least squares $f(\beta) = \frac{1}{2}\|X\beta - y\|^2$ with X of full column rank, identify the rate in terms of $\kappa(X)$.

Solution. [Supplement]

(a) The right side of the strong convexity inequality is minimized over y at $y - x = -\nabla f(x)/\mu$, giving $f(y) \ge f(x) - \|\nabla f(x)\|^2/(2\mu)$ for all y; take $y = x^\star$.

(b) As for Theorem 7.3, with $x^+ = x - \frac{1}{L}\nabla f(x)$,

$$f(x^+) \le f(x) - \frac{1}{2L}\|\nabla f(x)\|^2 \le f(x) - \frac{\mu}{L}\{f(x) - p^\star\}$$

by (a); subtract $p^\star$ and induct. Here $\mu \le L$ (compare the lower bound with the quadratic upper bound at $y \ne x$), so $0 \le 1 - \mu/L < 1$.

(c) $\nabla^2 f = X^\top X$ has eigenvalues in $[\sigma_{\min}(X)^2, \sigma_{\max}(X)^2]$; the exact quadratic Taylor expansion gives $\mu = \sigma_{\min}(X)^2$, $L = \sigma_{\max}(X)^2$. The rate is $1 - 1/\kappa(X)^2$: about $\kappa(X)^2$ iterations per constant-factor reduction. ■

Exercise 7.2 (Backtracking terminates)

Let ∇f be L-Lipschitz, $\nabla f(x) \ne 0$, $\Delta x = -\nabla f(x)$, $\alpha \in (0, \frac{1}{2}]$ and $\beta \in (0, 1)$.
(a) Show that the Armijo condition $f(x + s\Delta x) \le f(x) + \alpha s \nabla f(x)^\top \Delta x$ holds for every $s \in (0, 1/L]$.
(b) Deduce that backtracking from $s = 1$ stops after finitely many reductions, at some $s \ge s_{\min} = \min\{1, \beta/L\}$, and that f then decreases by at least $\alpha s_{\min}\|\nabla f(x)\|^2$.
(c) If f is bounded below, show that gradient descent with backtracking satisfies $\nabla f(x^{(t)}) \to 0$, without any convexity assumption.

Solution. [Supplement] Write $g = \nabla f(x)$.

(a) $f(x - sg) \le f(x) - s(1 - \frac{Ls}{2})\|g\|^2$, and for $s \le 1/L$, $1 - \frac{Ls}{2} \ge \frac{1}{2} \ge \alpha$, so $f(x - sg) \le f(x) + \alpha s\, g^\top \Delta x$.

(b) Trial steps $1, \beta, \beta^2, \dots$ reach $\beta^j \le 1/L$, so by (a) one is accepted. Either $s = 1$, or $s = \beta^j$ ($j \ge 1$) follows a rejected $\beta^{j-1} > 1/L$, so $s > \beta/L$. Either way $s \ge s_{\min}$ and $f(x^+) \le f(x) - \alpha s_{\min}\|g\|^2$.

(c) Summing (b), $\alpha s_{\min} \sum_{t<T} \|\nabla f(x^{(t)})\|^2 \le f(x^{(0)}) - \inf f < \infty$, so the terms tend to 0. ■

Exercise 7.3 (Coordinate descent for least squares is Gauss–Seidel)

Let $f(\beta) = \frac{1}{2}\|y - X\beta\|^2$ with columns $x_1, \dots, x_p$ of X all nonzero, and $G = X^\top X$.
(a) Show that one cycle of coordinate descent is exactly one Gauss–Seidel sweep for $G\beta = X^\top y$.
(b) Deduce that coordinate descent converges to the least squares solution when X has full column rank.
(c) Show that a cycle can be carried out in $O(np)$ flops without forming G.

Solution. [Supplement]

(a) In β_j alone, f is quadratic with leading coefficient $\frac{1}{2}\|x_j\|^2 > 0$ and $\partial f/\partial \beta_j = (G\beta)_j - (X^\top y)_j$, minimized by

$$\beta_j \leftarrow \frac{1}{g_{jj}}\Big((X^\top y)_j - \sum_{l \ne j} g_{jl}\beta_l\Big),$$

with β_l, $l < j$, already updated and $l > j$ old: the Gauss–Seidel update for $G\beta = X^\top y$.

(b) G is positive definite, so Exercise 5.3 gives convergence to $G^{-1}X^\top y = \hat{\beta}$.

(c) Keep $r = y - X\beta$; as $x_j^\top r = (X^\top y)_j - \sum_l g_{jl}\beta_l$, the update is

$$\beta_j \leftarrow \beta_j + \frac{x_j^\top r}{\|x_j\|^2}, \qquad r \leftarrow r - x_j\,\Delta\beta_j,$$

$O(n)$ per coordinate ($\|x_j\|^2$ precomputed), $O(np)$ per cycle, versus $O(np^2)$ to form G. ■